

NODE 5 — UPPER-BODY PHYSIOLOGIC CONTEXT DEVICE

A. Overview

In certain embodiments, a fifth node is configured as an upper-body physiologic context device configured to measure physiologic signals that improve interpretation of intimate-region patterns by providing context regarding cardiac dynamics, autonomic state, respiration proxies, activity state, thermal state, and wear integrity. The fifth node may be implemented as a chest-mounted device, a ring, a wrist-worn device, or other wearable form factor. Node 5 may be used during sleep and/or daily activity and may provide longitudinal context suitable for cross-device correlation with one or more other nodes.

B. Form Factor and Wear Locations

The fifth node may be implemented in any of multiple form factors, including: 1. a chest-mounted device configured to maintain electrode contact and stable mechanical coupling; 2. a ring configured for finger wear with optical coupling for pulse sensing and stable inertial sensing; 3. a wrist-worn device configured for extended wear and multi-sensor integration; and/or 4. an adhesive patch or other attachment format configured for enhanced contact integrity. In certain embodiments, the device includes one or more contact surfaces, electrode placement features, strap or band structures, compliance regions, and/or contact-integrity sensing features to ensure signal quality during movement.

C. Sensor Suite

The fifth node may include one or more sensors and sensing elements, including combinations of the following: 1. Electrical cardiac sensing using ECG electrodes to capture cardiac electrical activity and derived metrics including heart rate and heart rate variability. 2. Optical pulse sensing including PPG sensing to capture pulse characteristics and to provide redundancy or cross-validation with ECG-derived measures. 3. Inertial sensing including a 6-axis IMU to capture activity state, posture proxies, motion gating, and artifact rejection. 4. Respiratory dynamics sensing or respiratory proxy sensing, including impedance-derived respiration, accelerometer-derived respiration proxies, ECG-derived respiration, acoustic proxies, or other suitable approaches. 5. Temperature or thermal context sensing to capture skin temperature and/or ambient temperature and to support physiologic interpretation and wear detection. 6. Electrodermal activity sensing and/or skin conductance sensing to capture autonomic proxies and stress-related signatures. 7. Contact integrity and wear detection sensing, including impedance contact checks, electrode integrity checks, optical coupling checks, pressure/contact checks, and/or capacitive proximity/contact confirmation.

D. Data Collection, Wear Detection, and Signal Quality

The fifth node may operate continuously, periodically, or in event-triggered sampling modes. Wear detection may be performed using contact integrity sensing, temperature dynamics, electrode impedance checks, optical coupling checks, and/or multi-sensor agreement. The system may compute a signal quality index to identify segments affected by motion, poor contact, sweating, electrode lift, optical decoupling, or environmental changes. In certain embodiments, inertial sensing is used for motion gating, and contact integrity sensing is used to validate stable measurement windows.

E. Derived Features and Metrics

The computing system may derive features from Node 5 signals including heart rate measures, HRV measures, respiratory proxy measures, activity state classification, posture proxies, temperature context features, electrodermal features, sleep-state proxies, and confidence measures. Metrics may be presented as privacy-preserving outputs including indices, zones, ranges, trend curves, and confidence indicators. Example outputs include: a cardiac context index derived from ECG and/or PPG features, an autonomic proxy index derived from HRV and/or electrodermal activity, an activity or motion state label and associated confidence indicator, a respiratory trend curve or respiratory proxy metric, and a wear-integrity score based on contact and signal quality.

F. Cross-Device Correlation and Conditioning of Interpretation

In various embodiments, Node 5 outputs are used to condition interpretation of intimate-region patterns from other nodes. For example, intimate-region metrics may be interpreted differently based on activity state, sleep state, respiratory dynamics, thermal context, or autonomic proxies. Node 5 may support disambiguation between physiologic change and artifact, including identifying whether apparent changes coincide with motion state, poor contact, stress proxies, or temperature shifts. In certain embodiments, Node 5 provides timestamps, event markers, and context labels used to segment and interpret data from other nodes.

G. Variations and Alternatives

The fifth node may be implemented with different sensor combinations. Certain embodiments emphasize ECG with HRV; other embodiments emphasize PPG with inertial sensing; other embodiments include electrodermal sensing. The device may include local storage, intermittent upload, or real-time streaming. Power may be provided via rechargeable battery, inductive charging, or other power subsystems. Processing may be performed locally or via a computing system.